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JOINING JOINT TO BE JOINED TO RESIN, METHOD FOR MANUFACTURING JOINING JOINT, AND JOINING STRUCTURE BETWEEN METAL AND RESIN

Abstract

Provided is a joining structure between metal and resin which has improved junction strength. A joining joint has a hand-drum shape having a constricted portion constricted between both of the bottom surfaces by forming a groove-shaped recess in a circumferential direction of a substantially cylindrical side surface made of metal. Upper and lower end portions of the side surface and the recess form a rough concavo-convex shape, and the side surface including the recess becomes a junction surface. On the junction surface, a fine concavo-convex shape with arithmetic average roughness Ra of 10 to 300 μm is formed. By forming the rough concavo-convex shape and the fine concavo-convex shape on the junction surface, the joining joint makes strong joining to resin.

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Background/Summary

TECHNICAL FIELD

[0001] The technical field of the present description relates to a joining structure between metal and resin.

BACKGROUND ART

[0002] Conventionally, a joining structure between metal and resin has been known. In a joining structure between metal and resin described in Patent Literature 1, a metal surface is knurled, and resin is joined to knurl made by knurling. In a bearing device as a joining structure between metal and resin described in Patent Literature 2, a bearing portion and a bolt hole forming portion are made of metal, and a bracket connecting the bearing portion and the bolt hole forming portion is made of resin. Accordingly, the bearing device is made light in weight. Patent Literature 2 is a patent application filed by the applicant of the present application.

CITATION LIST

Patent Literatures

[0003] Patent Literature 1: JP No. 2018-115679 A

[0004] Patent Literature 2: JP No. 2020-012502 A

SUMMARY OF INVENTION

Technical Problem

[0005] Concerning the conventional joining structures between metal and resin described in these Patent Literatures, an improvement in junction strength between metal and resin has been desired.

[0006] An object to be achieved by the technique described in the present application is in view of these respects described above, and is to provide a joining structure between metal and resin which has improved junction strength.

Solution to Problem

[0007] A joining joint to be joined to resin according to an embodiment described in the present description is made of metal, has a junction surface to be joined to resin on a surface, and has a fine concavo-convex shape formed on the junction surface, where arithmetic average roughness Ra of the fine concavo-convex shape is 10 to 300 μm .

[0008] According to the joining joint to be joined to resin of an embodiment described in the present description, an area of contact between the junction surface and the resin can be increased, and resin intruding into the fine concavo-convex shape of the junction surface exerts an effect as an anchor, so that the junction strength between metal and resin can be improved.

[0009] Here, in the above-described joining joint, it is also possible that a groove-shaped recess is formed on the junction surface.

[0010] According to this, a rough concavo-convex shape is formed on the junction surface due to the groove-shaped recess, and resin intrudes into the rough concavo-convex shape, so that the junction strength between metal and resin can be improved.

[0011] It is also possible that, in the joining joint described above, the junction surface has a substantially cylindrical shape serving as a side surface, and the recess is formed in a circumferential direction of the side surface.

[0012] According to this, resin can be joined to the substantially cylindrical joining joint across a circumferential direction of the side surface.

[0013] In the joining joint, it is possible that a cross section in a radial direction of a constricted

portion formed of the recess in the circumferential direction is substantially polygonal.

[0014] According to this, the constricted portion formed of the recess and the resin intruding into the recess fit each other in a substantially polygonal shape, so that the junction strength between metal and resin with respect to rotation of the joining joint in the circumferential direction can be improved.

[0015] Here, a method for manufacturing a joining joint according to an embodiment described in the present description is a method for manufacturing the joining joint described above, wherein [0016] a fine concavo-convex shape is formed on a surface of a casting mold for casting the joining joint by forming the casting mold of coarse aggregate, and [0017] a fine concavo-convex shape can be formed on a surface of the joining joint casted from the casting mold.

[0018] According to the method for manufacturing a joining joint according to an embodiment described in the present description, a casting mold having a fine concavo-convex shape on a surface is formed of aggregate with a coarse grain size, and therefore, the fine concavo-convex shape can be inexpensively formed.

[0019] In the above-described method for manufacturing the joining joint, it is possible that an average grain size (median diameter d_{50}) of the aggregate is 200 to 2000 μm .

[0020] According to this, a fine concavo-convex shape that improves the junction strength to resin can be formed on a junction surface.

[0021] Here, a joining structure between metal and resin according to an embodiment described in the present description includes: [0022] the joining joint, and [0023] a resin bracket that covers the junction surface of the joining joint, and is formed of resin for fixing the joining joint as a matrix.

[0024] According to the joining structure between metal and resin according to the embodiment described in the present description, the joining joint and the resin bracket can be strongly joined.

Advantageous Effect of Invention

[0025] According to a joining joint of an embodiment described in the present description, the junction strength between metal and resin can be improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0026] FIG. 1 is a perspective view of a joining joint according to an embodiment.

[0027] FIG. 2 is a front view of the same joining joint.

[0028] FIG. 3 is a sectional view taken along line III-III in FIG. 2.

[0029] FIG. 4 is a sectional view of a casting mold for casting the same joining joint.

[0030] FIG. 5 is a perspective view of a joining structure using the same joining joint.

[0031] FIG. 6 is a front view of a bearing device using the same joining joint.

[0032] FIG. 7 is a plan view of the same bearing device.

DESCRIPTION OF EMBODIMENTS

[0033] Hereinafter, joining joints to be joined to resin according to embodiments described in the present description will be described based on the drawings. However, the technique described in this description is not limited to these embodiments. A joining joint **1** according to an embodiment is used as an insert member (insert collar, insert nut), and as illustrated in FIG. 1, the joining joint is formed into a hand-drum shape having a groove-shaped recess **13** formed in a circumferential direction of a substantially cylindrical side surface **11** of metal, and a constricted portion **14** constricted between both bottom surfaces **12**. A rough concavo-convex shape **101** is formed of upper and lower end portions **15** of the side surface **11** and the recess **13**, and the side surface **11** including the recess **13** becomes a junction surface **10** to be joined to resin. On the junction surface **10**, a fine concavo-convex shape **100** with arithmetic average roughness R_a of 10 to 300 μm is formed. Due to the formation of the rough concavo-convex shape **101** and the fine concavo-convex

shape **100** on the junction surface **10**, the joining joint **1** can be strongly joined to resin. In addition, as illustrated in FIGS. **6** and **7**, the joining joint **1** of the embodiment can be used as a bolt hole forming portion of a bearing device in which a shaft bearing portion **201** and a bolt hole forming portion (joining joint **1**) are formed of metal, and a bracket (resin bracket **2**) connecting the shaft bearing portion **201** and the bolt hole forming portion is formed of resin as described in Patent Literature 2 (Japanese Published Unexamined Patent Application No. 2020-012502) described above. Accordingly, the bearing device as a joining structure between metal and resin can be made light in weight. In this description, concerning orientation of the joining joint **1**, as illustrated in FIG. **1**, a direction of an axis connecting the centers of both of the bottom surfaces **12** is set as the up-down direction, and in the illustration, U represents up, and D represents down. In addition, a radial direction and a circumferential direction may be expressed with respect to the axis connecting the centers of each of the bottom surfaces **12**.

[0034] The joining joint **1** is formed of metal, and as the metal, specifically, a nodular graphite cast iron product FCD500 (JIS G 5502:2007) is used. The nodular graphite cast iron product has excellent tensile strength and heat resistance, and excellent durability as the joining joint **1**. As another embodiment, aluminum alloy diecasting ADC12 (JIS H 5302:2006) is used. ADC12 is excellent in mechanical properties, machinability, and castability. The joining joint **1** is manufactured by die-casting using a casting mold which realizes low-cost manufacturing and is suitable for mass production although details thereof will be described later.

[0035] As illustrated in FIGS. **1** to **3**, the joining joint **1** of the embodiment has a hand-drum shape including one groove-shaped recess **13** formed across the whole circumference of the substantially cylindrical side surface **11** in the circumferential direction, thick upper and lower end portions **15** and **15**, and the constricted portion **14** at the center in the up-down direction. The joining joint **1** has a rough concavo-convex shape **101** formed of the upper and lower end portions **15** and **15** and the constricted portion **14** formed at the center in the up-down direction. Due to the rough concavo-convex shape **101**, as illustrated in FIG. **5**, when the joining joint **1** is fixed to the resin bracket **2**, resin intrudes into the recess, and the effect of preventing the joining joint **1** from coming off is improved.

[0036] As illustrated in FIG. **2**, at external corner portions in the circumferential direction which define the recess **13**, chamfered portions **13a** chamfered to be round inward are formed, and at internal corner portions in the circumferential direction, chamfered portions **13b** chamfered to be round outward are formed. Due to the chamfered portions **13a** and the chamfered portions **13b**, a stress to be applied to resin when the joining joint **1** is fixed to the resin bracket **2** can be distributed.

[0037] As illustrated in FIG. **3**, the constricted portion **14** has a substantially quadrangular cross section in a radial direction. Since the constricted portion **14** has a substantially quadrangular cross section, when the joining joint **1** is fixed to the resin bracket **2**, the constricted portion **14** formed of the recess **13** and the resin that has intruded into the recess **13** fit in a quadrangular shape, so that the junction strength between metal and the resin with respect to rotation of the joining joint in the circumferential direction can be improved. At external corner portions in the up-down direction which form the substantially quadrangular shape of the constricted portion **14**, chamfered portions **14a** chamfered to be round inward are formed. Due to the chamfered portions **14a**, a stress to be applied to resin when the joining joint **1** is fixed to the resin bracket **2** can be distributed.

[0038] Each of upper and lower bottom surfaces **12** of the joining joint **1** has a circular shape, and has a screw hole **16** opened through a central axis in the up-down direction. Due to the screw hole **16**, the joining joint **1** can be joined to another member by a screw or the like. The screw hole **16** may be changed to a threadless circular or odd-shaped hole such as a polygonal hole.

[0039] The side surface **11** including the recess **13** of the joining joint **1** of the embodiment serves as a junction surface **10** to be joined to resin of the resin bracket **2**. On the junction surface **10**, a fine concavo-convex shape **100** with arithmetic average roughness Ra of approximately 90 μm is

formed. Due to the formation of the fine concavo-convex shape **100** with arithmetic average roughness Ra of approximately 90 μm on the junction surface **10**, when the joining joint **1** is joined to resin of the resin bracket **2**, the resin intrudes into the fine concavo-convex shape **100** and exerts an effect as an anchor, so that coupling between the joining joint **1** and the resin can be made strong.

[0040] Although details of the fine concavo-convex shape **100** with arithmetic average roughness Ra of approximately 90 μm will be described later, a casting mold **3** for casting the joining joint **1** is formed of coarse aggregate **32** (FIG. 4), and accordingly, the fine concavo-convex shape **100** is formed on a surface of the casting mold **3**, and is formed by transferring the fine concavo-convex shape **100** onto the side surface **11** of the joining joint **1** casted from the casting mold **3**.

[0041] Next, a method for manufacturing the joining joint **1** of the embodiment will be described. The joining joint **1** is manufactured by casting using the casting mold **3**, and the method for manufacturing the joining joint **1** of the embodiment will be described by being divided into a method for manufacturing the casting mold **3** for casting the joining joint **1** and a method for manufacturing the joining joint **1** by using the casting mold **3**.

[0042] The method for manufacturing the casting mold **3** includes a primary mold manufacturing process for manufacturing a primary mold by replicating a master model from a master model (3D data) of the joining joint **1**, and a casting mold manufacturing process for manufacturing a secondary mold (casting mold **3**) having the fine concavo-convex shape **100** on a surface from the primary mold by using coarse aggregate **32**.

[0043] In the primary mold manufacturing process, by a versatile method, a primary mold made of metal (or resin) is manufactured by replicating a master model from a master model of the joining joint **1**.

[0044] In the casting mold manufacturing process, as illustrated in FIG. 4, the casting mold **3** (secondary mold) is molded by pouring a die molding material for forming a gas-hardening sand mold so as to make a cast of the primary mold. The die molding material for forming a gas-hardening sand mold is hardened by using a cold box process as a kind of gas-hardening mold making methods in which the die molding material is hardened by supplying gas thereto, and accordingly, the casting mold **3** is molded.

[0045] The die molding material is prepared by mixing aggregate **32**, resin, and a hardening agent, and as the aggregate **32**, molding silica sand #3.5 (JIS G 5901:2016) as aggregate which remains most on a screening surface of a sieve with a nominal opening of 1180 μm is used, and as the resin and hardening agent, a general-purpose resin and hardening agent to be used for the cold box process are used. The die molding material contains 98.4% aggregate **32**, 0.8% resin, and 0.8% hardening agent.

[0046] The die molding material is mostly made of coarse aggregate **32**, so that the casting mold **3** made of the die molding material has, as a surface shape, a fine concavo-convex shape **100** formed of the shape (grain size) of the aggregate **32**.

[0047] A method for manufacturing the joining joint **1** by using the casting mold **3** consists of a pouring process of pouring molten metal into the casting mold **3**, and a demolding process of demolding to take the joining joint **1** out.

[0048] In the pouring process, molten metal is poured into the casting mold **3**, and cooled and solidified to mold a joining joint **1**. Pouring is performed by a versatile method.

[0049] In the demolding process, the joining joint **1** is demolded from the casting mold **3**, unnecessary objects such as burrs are removed, and by machining the bottom surfaces **12**, the fine concavo-convex shapes **100** of the bottom surfaces **12** are eliminated, and the joining joint **1** is made usable.

[0050] In the joining joint **1** manufactured in this way, on the junction surface **10** of the side surface **11**, the fine concavo-convex shape **100** is formed. In addition, the rough concavo-convex shape **101** is formed of the upper and lower end portions **15** and **15**, the groove-shaped recess **13** in the

circumferential direction of the side surface **11** of the joining joint **1**, and the constricted portion **14** at the center in the up-down direction. In the joining joint **1**, just by changing the size of the aggregate **32** forming the casting mold **3**, the size of the fine concavo-convex shape **100** can be adjusted.

[0051] Next, a joining structure between metal and resin which uses the joining joint **1** of the embodiment will be described. In the embodiment, concerning a bearing device that supports a shaft of a vehicle turnably, a joining structure using the joining joint **1** at the bolt hole forming portion to be joined to a vehicle body will be described by way of example.

[0052] As illustrated in FIGS. **6** and **7**, the bearing device **200** includes an annular shaft bearing portion **201** through which a shaft is inserted via a bearing, joining joints **1** into which bolts for fixing the shaft bearing portion **201** to a fixation portion, and a bracket that connects the shaft bearing portion **201** and the joining joints **1**, the shaft bearing portion **201** and the joining joints **1** are formed of metal, and the bracket (resin bracket **2**) connecting the shaft bearing portion **201** and the joining joints **1** are formed of resin. Accordingly, the bearing device **200** is made light in weight.

[0053] The resin bracket **2** integrally connects the shaft bearing portion **201** and the joining joints **1**, and is made of a resin mixture, and molded by a forming mold. As resin to be used in the resin mixture, two types of resins are used separately. One of the resins is polyamide resin as thermoplastic resin, and another resin is phenol resin as thermosetting resin. By using polyamide resin or phenol resin as a resin to be used in the resin mixture, the resin bracket **2** can be made excellent in heat resistance and mechanical strength.

[0054] Glass fiber is mixed as fiber into the resin mixture to be used for the resin bracket **2**, and accordingly made as a fiber-mixed resin. By mixing fiber into the resin mixture, hardening shrinkage of the resin can be reduced, and the physical strength can be improved. A mixture fraction of fiber in the fiber-mixed resin is set to 40 mass % to efficiently improve the physical strength.

[0055] The joining joint **1** used in the bearing device **200** has the fine concavo-convex shape **100** formed on the junction surface **10** of the side surface **11** including the recess **13**, so that joining to the resin of the resin bracket **2** can be made strong. As illustrated in FIG. **5**, the joining joint **1** has the rough concavo-convex shape **101** formed of the upper and lower end portions **15** and **15** and the recess **13** at the center in the up-down direction, so that the effect of preventing coming-off from the resin bracket **2** is improved, and since the cross section of the constricted portion **14** in a radial direction is substantially quadrangular, the rotation prevention effect is improved.

[0056] The joining joint **1** and the connection structure between metal and resin of the embodiment can also be realized when their configurations are changed to the following forms.

[0057] Although the nodular graphite cast iron product FCD500 is used as an iron alloy (cast iron) used for molding the joining joint **1** in the joining joint **1** of the embodiment, nodular graphite cast iron products (FCD350, FCD400, FCD450, FCD600, FCD700, FCD800) prescribed in JIS G 5502:2001 may be used. This is because these are excellent in mechanical strength.

[0058] Although ADC12 is used as aluminum alloy diecasting for molding the joining joint **1** in the joining joint **1** of the embodiment, aluminum alloy diecastings (ADC1, ADC3, ADC5, ADC6, ADC10, ADC10Z, ADC12Z, ADC14) prescribed in JIS H 5302:2006 may be used.

[0059] In the joining joint **1** of the embodiment, although the cross section of the constricted portion **14** in a radial direction is set to be substantially quadrangular, the cross-sectional shape of the constricted portion **14** in a radial direction may be set to be substantially triangular to substantially hexagonal. This improves the coming-off prevention effect and the rotation prevention effect with respect to the resin bracket **2**. The cross section of the constricted portion **14** in a radial direction may be set to be circular.

[0060] In the joining joint **1** of the embodiment, one groove-shaped recess **13** is provided across the whole circumference of the side surface **11** in the circumferential direction, however, one to three

groove-shaped recesses **13** may be provided in the up-down direction. This improves the coming-off prevention effect with respect to the resin bracket **2**. When groove-shaped recesses **13** more than three in number are provided, the structure of the joining joint **1** may become complicated. As another embodiment, one or two groove-shaped recesses **13** may be provided in the up-down direction.

[0061] The joining joint **1** of the embodiment is a substantially cylindrical insert member (insert collar, insert nut), however, without limitation to the insert member, joining joints to be used for joining metal and resin may be used. The shape is not limited to the substantially cylindrical shape, and may be, for example, a conical shape, a prismatic shape, a pyramidal shape, etc.

[0062] In the method for manufacturing the joining joint **1** of the embodiment, the casting mold **3** is molded by a gas-hardening sand mold, however, as the casting mold **3**, a green sand mold, a self-hardening sand mold, a thermosetting sand mold, etc., may also be used for molding.

[0063] In the method for manufacturing the joining joint **1** of the embodiment, as the aggregate **32** to be mixed into the die molding material for forming the gas-hardening sand mold serving as the casting mold **3**, molding silica sand **190 3** (average grain size (median diameter d_{50}): $1500\ \mu\text{m}$) as aggregate which remains most on a screening surface of a sieve with a nominal opening of $1180\ \mu\text{m}$ is used, however, as the aggregate **32**, aggregate may be used as long as its average grain size is 200 to $2000\ \mu\text{m}$. This enables formation of the fine concavo-convex shape **100** that makes strong joining to resin. When the average grain size of the aggregate is less than $200\ \mu\text{m}$, the fine concavo-convex shape **100** becomes finer, and joining to resin may not be made strong. On the other hand, when the average grain size is more than $2000\ \mu\text{m}$, the strength of the casting mold **3** may deteriorate. As another embodiment, the average grain size of the aggregate **32** may be 600 to $1700\ \mu\text{m}$.

[0064] In the method for manufacturing the joining joint **1** of the embodiment, as the aggregate **32** to be mixed into the die molding material for forming the gas-hardening sand mold serving as the casting mold **3**, molding silica sand **#3** as aggregate which remains most on a screening surface of a sieve with a nominal opening of $1180\ \mu\text{m}$ is used, however, as the silica sand, molding silica sand **#6** to molding silica sand **#3** may be used. As another embodiment, the size of silica sand may be molding silica sand **#4** to molding silica sand **#3**. As the aggregate **32**, calcium carbonate powder, cerben (crushed sanitary ware), crushed stone powder, etc., may also be used as long as they have an average grain size equivalent to the size described above.

[0065] In the method for manufacturing the joining joint **1** of the embodiment, as the die molding material for forming the gas-hardening sand mold serving as the casting mold **3**, a die molding material containing 98.4% aggregate **32**, 0.8% resin, and 0.8% hardening agent is used, however, a die molding material may be used as long as it contains 70 to 99% aggregate **32**, 0.5 to 15% resin, and 0.5 to 15% hardening agent. As another embodiment, the die molding material may contain 90 to 99% aggregate **32**, 0.5 to 5% resin, and 0.5 to 5% hardening agent.

[0066] As the thermoplastic resin of the resin mixture for forming the resin bracket **2**, polyamide resin has been used in the embodiment, however, thermoplastic resin such as polypropylene, polyvinyl chloride, polystyrene, acrylic resin, polycarbonate resin, ABS resin may be used which have excellent heat resistance.

[0067] As the thermosetting resin of the resin mixture for forming the resin bracket **2**, phenol resin has been used in the embodiment, however, thermosetting resin such as amino resin and urea resin may be used. This is because these have excellent heat resistance.

[0068] As fiber of the fiber-mixed resin as the resin mixture for forming the resin bracket **2**, glass fiber that is inorganic fiber is used in the embodiment, however, inorganic fiber such as carbon fiber, metal fiber, and ceramic fiber, or organic fiber such as cellulose fiber, acrylic fiber, polyamide fiber, and polyester fiber may also be used.

[0069] The mixture fraction of fiber in the fiber-mixed resin is 40 mass % in the embodiment, however, fiber-mixed resin may be used as long as the mixture fraction of fiber is 10 to 70 mass %.

A mixture fraction less than 10 mass % may not sufficiently improve the strength. On the other hand, a mixture fraction more than 70 mass % may cause an increase in density of the fiber-mixed resin and may not make the bearing device 200 light in weight. The mixture fraction is more preferably 20 to 60 mass %, and further preferably 30 to 50 mass %.

[0070] The fiber-mixed resin containing polyamide resin or phenol resin as resin and mixed with 40 mass % glass fiber is used in the embodiment, however, fiber-mixed resin available on the market may also be used. Usable glass-fiber-mixed resin available on the market includes A1022GFL15, A1022GFL, A1030GFL, A1030GFL45, A1022GFL60 (A), EX-8406G30, A175S, A190S, A192S, A690S (the foregoing resins are made by UNITIKA, LTD.), CM1011G-15, CM1001G-15, CM1001G-20, CM1011G-30, CM1016G-30, CM1011G-45, CM1012G-45N, CM3001G-15, CM3006G-15, CM3001G-30, CM3006G-30, CM3001G-45, CM3006G-45 (the foregoing resins are made by Toray Industries, Inc.), etc.

REFERENCE SIGNS LIST

[0071] **1** Joining joint [0072] **2** Resin bracket [0073] **3** Casting mold [0074] **3a** Side surface portion [0075] **10** Junction surface [0076] **11** Side surface [0077] **12** Bottom surface [0078] **13** Recess [0079] **13a** Chamfered portion [0080] **13b** Chamfered portion [0081] **14** Constricted portion [0082] **14a** Chamfered portion [0083] **15** End portion [0084] **16** Screw hole [0085] **32** Aggregate [0086] **100** Fine concavo-convex shape [0087] **101** Rough concavo-convex shape [0088] **200** Bearing device [0089] **201** Shaft bearing portion

Claims

1. A joining joint made of metal, having a junction surface to be joined to resin on a surface, and having a fine concavo-convex shape formed on the junction surface, where arithmetic average roughness Ra of the fine concavo-convex shape is 10 to 300 μm .
 2. The joining joint according to claim 1, wherein a groove-shaped recess is formed on the junction surface.
 3. The joining joint according to claim 2, wherein the junction surface has a substantially cylindrical shape serving as a side surface, and the recess is formed in a circumferential direction of the side surface.
 4. The joining joint according to claim 3, wherein a cross section in a radial direction of a constricted portion formed of the recess in the circumferential direction is substantially polygonal.
 5. A method for manufacturing the joining joint according to claim 1, wherein a fine concavo-convex shape is formed on a surface of a casting mold for casting the joining joint by forming the casting mold of coarse aggregate, and a fine concavo-convex shape is formed on a surface of the joining joint casted from the casting mold.
 6. The method for manufacturing the joining joint according to claim 5, wherein an average grain size (median diameter d50) of the aggregate is 200 to 2000 μm .
 7. A joining structure between metal and resin, comprising: the joining joint according to claim 1; and a resin bracket that covers the junction surface of the joining joint, and is formed of resin for fixing the joining joint as a matrix.
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